

# Qingyuan YAO

(aka. Game/App/Web Developer)

## Education

|                |                                                                                                                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apr - Jul 2024 | <b>POEC Training for .Net Developer</b><br>Global Knowledge Network                                                                                                                                |
| 2021 - 2023    | <b>Master of Artificial Intelligence</b><br><i>Parcours AgeNts Distribués, Robotique, Recherche Opérationnelle, Interaction, DECision (ANDROIDE)</i><br><b>Mention Bien</b><br>Sorbonne University |
| 2018 - 2021    | <b>Bachelor of Computer Science</b><br><b>Mention Bien</b><br>Sorbonne University                                                                                                                  |

## Portfolio



📍 Ile-de-France (relocatable)

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## SOFT SKILLS

**Creative:** Full of ideas, passionate in many areas

**Adaptive:** Lived in 3 countries, fast learner

**Collaborative:** Motivated by teammates, a good team player in 3 time-zones, good at being the mediator

**Open-minded:** Love cultural & sociological differences

**Empathetic:** Sensitive to other's experience

## Hard Skills

- **Programming:** Python, C, C#, C++, Java, Javascript, LaTeX
- **Game Development**
  - **Unity (Experienced):** Desktop, Mobile, Web, etc.
  - **UE5 (Beginner):** Desktop, C++, Blueprint
- **Development:** MAUI, ASP.Net, Azure, Firebase, SQL, OOP, UML, HTML, CSS, React, Next.js, Tailwind
- **Research:** Serious Games, User Tracing/Modeling/Adaptation, HCI, Deep/Reinforcement Learning, Multi-Agent System, Game Theory, Operational Research, Decision
- **Languages:** Chinese (native), English (fluent), French (fluent), Spanish (limited), Japanese (limited)
- **Art:** Digital & Analog Photography, Video Montage, Singing, Songwriting, Music Production

## Projects

### Ongoing Projects

#Web #JavaScript #React #Next.js #C# #MAUI #iOS #Android #Unity

**Forêt Hubert Reeves:** A memorial website for astrophysicist Hubert Reeves with locations of planted ginkgo trees

**Wa-Nai-Sa:** Portfolio website for an artist

**KaiSeki :** An app that uses LLM to analyse Japanese sentences

**PastelDices:** A game jam project polished for publishing

### Game Projects

#Unity #UnrealEngine #HCI #\$\_recognizer #EntityComponentSystem  
#CbKST #BayesianKnowledgeTracing #xAPI

**Zhuiyi - In Search of Lost Words:** A serious game that teaches to write Chinese characters with Wacom graphic tablet and a Nintendo Switch Joycon

**SPY:** Enhancement of a game that teaches programming

**Game Jams** (cf. [Game Portfolio](#))

### AI Projects

#Python #TensorFlow #DeepLearning #LSTM #A2C #Multi-Agent #Java

**Meta-RL Project:** Reproduction and analysis of the paper [Prefrontal Cortex as a Meta-Reinforcement Learning System](#)

**Multi-Agent Simulation of Ecotopia:** Simulation of a fictional society based on the book Ecotopia

**Multi-Agent Project Dedale:** Multi-agent system where agents work together to collect treasures

### HCI Project

**Immersive Book:** Conception of a holographic interactive book with user-centered design while following HCI protocols

**Gestures vs. Keyboard:** Data collection and comparison of gestural and keyboard shortcuts under HCI protocols

## Experience

### Freelance App Developer

Arborus

Ongoing

#C# #MAUI #iOS #Android

Development of an app that fights sexism with comebacks.

### Deep Generative AI Internship

Tokyo National Institute of Informatics

2023

Mar. - Aug.

#Python #PyTorch #DeepLearning

Creation of a new dataset of audio-text-motion using existing models for 3D pose estimation, and implementation of a [Transformer VAE](#) baseline model for proofing validity.

### Serious Game Development Internship

LIP6, Sorbonne University

2022

Jun. - Jul.

#Unity #C# #SeriousGame

- Solo developer of on the prototype targeted for PC and Android, developed by previous team of students
- Significantly improved stability and optimization by using Coroutine, async for asynchronous execution, as well as basic Jobs System and DOTS for multi-threading
- Added a visual novel interface with loadable conversation tree and a custom 2D [A\\* pathfinding](#) system